Experiment No: 4

Title: Implementation of Storage as a Service.

Aim: To study Implementation of Storage as a Service by using AWS.

Theory: Amazon EC2 provides you with flexible, cost effective, and easy-to-use data storage options for your instances. Each option has a unique combination of performance and durability. These storage options can be used independently or in combination to suit your requirements.

After reading this section, you should have a good understanding about how you can use the data storage options supported by Amazon EC2 to meet your specific requirements. These storage options include the following:

- Amazon Elastic Block Store
- Amazon EC2 instance store
- EFS
- S3

The following figure shows the relationship between these storage options and your instance.

Amazon EBS

Amazon EBS provides durable, block-level storage volumes that you can attach to a running instance. You can use Amazon EBS as a primary storage device for data that requires frequent and granular updates. For example, Amazon EBS is the recommended storage option when you run a database on an instance.

Amazon EC2 instance store

Many instances can access storage from disks that are physically attached to the host computer. This disk storage is referred to as *instance store*. Instance store provides temporary block-level storage for instances. The data on an instance store volume persists only during the life of the associated instance; if you stop, hibernate, or terminate an instance, any data on instance store volumes is lost

Amazon EFS file system

Amazon EFS provides scalable file storage for use with Amazon EC2. You can create an EFS file system and configure your instances to mount the file system. You can use an EFS file system as a common data source for workloads and applications running on multiple instances.

Amazon S3

Amazon S3 provides access to reliable and inexpensive data storage infrastructure. It is designed to make web-scale computing easier by enabling you to store and retrieve any amount of data, at any time, from within Amazon EC2 or anywhere on the web

Implementation of Storage as a Service by using AWS S3:

What is AWS S3?

Amazon Simple Storage Service (S3) is storage for the internet. It is designed for large-capacity, low-cost storage provision across multiple geographical regions. Amazon S3 provides developers and IT teams with Secure, Durable and Highly Scalable object storage.

S3 is Secure because AWS provides:

- Encryption to the data that you store. It can happen in two ways:
    - Client Side Encryption
    - Server Side Encryption
- Multiple copies are maintained to enable regeneration of data in case of data corruption
- *Versioning*, wherein each edit is archived for a potential retrieval.

S3 is Durable because:

- It regularly verifies the integrity of data stored using checksums e.g. if S3 detects there is any corruption in data, it is immediately repaired with the help of replicated data.
- Even while storing or retrieving data, it checks incoming network traffic for any corrupted data packets.

S3 is Highly Scalable, since it automatically scales your storage according to your requirement and you only pay for the storage you use.

What kind and how much of data one can store in AWS S3?

You can store virtually any kind of data, in any format, in S3 and when we talk about capacity, the volume and the number of objects that we can store in S3 are unlimited.

*An object is the fundamental entity in S3. It consists of data, key and metadata.

When we talk about data, it can be of two types-

- Data which is to be accessed frequently.
- Data which is accessed not that frequently.

Therefore, Amazon came up with 3 storage classes to provide its customers the best experience and at an affordable cost.

Let’s understand the 3 storage classes with a “health-care” use case:

1. Amazon S3 Standard for frequent data access
   This is suitable for performance sensitive use cases where the latency should be kept low. e.g. in a hospital, frequently accessed data will be the data of admitted patients, which should be retrieved quickly.

2. Amazon S3 Standard for infrequent data access

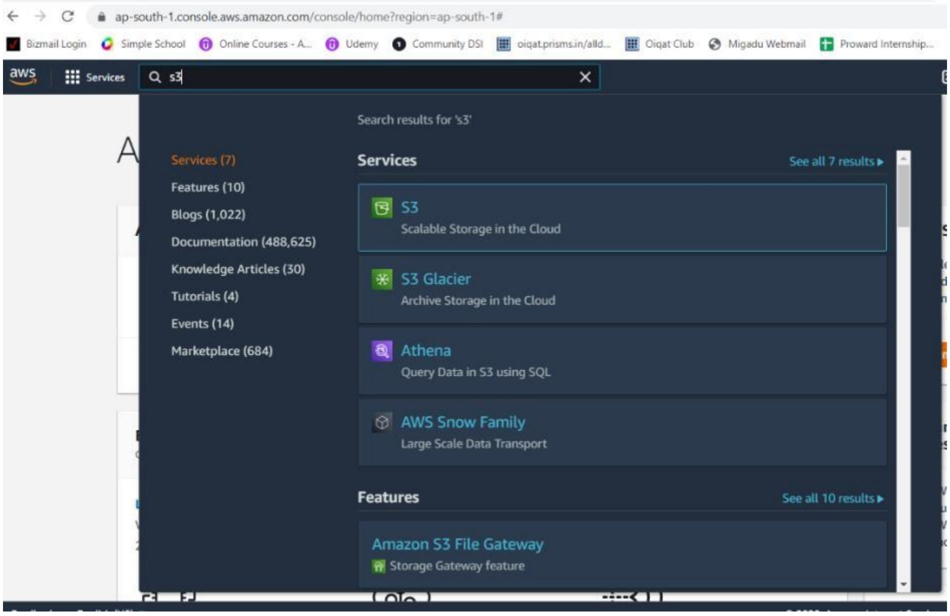
   This is suitable for use cases where the data is long lived and less frequently accessed,i.e for data archival but still expects high performance. e.g. in the same hospital, people who have been discharged, their records/data will not be needed on a daily basis, but if they return with any complication, their discharge summary should be retrieved quickly.

3. Amazon Glacier
   Suitable for use cases where the data is to be archived and high performance is not required, it has a lower cost than the other two services.e.g. in the hospital, patients’ test reports, prescriptions, MRI, X Ray, Scan docs etc. that are older than a year will not be needed in the daily run and even if it is required, lower latency is not needed.

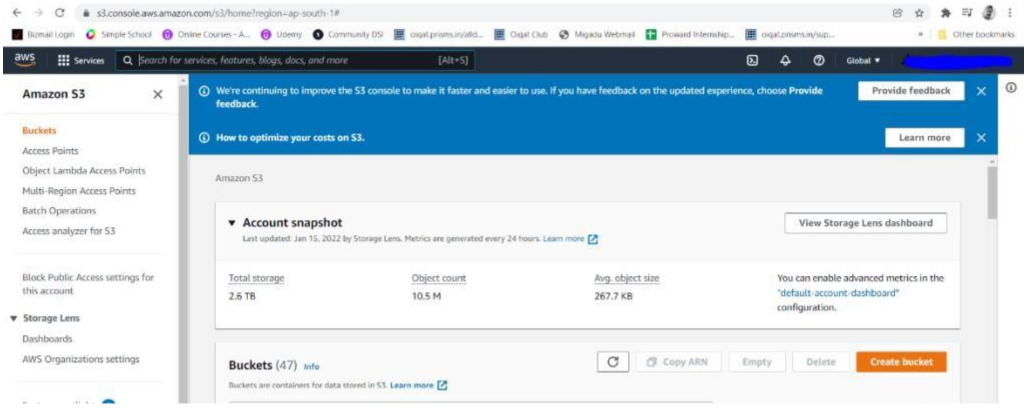
Characteristics	Standard	Standard - Infrequent Access	Glacier
Durability	99.99%	99.99%	99.99%
Availability	99.99%	99.90%	N/A
Minimum Object Size	No limit	128KB	No limit
Minimum Storage Duration	No minimum duration	30 Days	90 Days
First Byte Latency	milliseconds	milliseconds	4 hours
Retrieval Fee	No Fee	per GB retrieved	per GB retrieved

Output

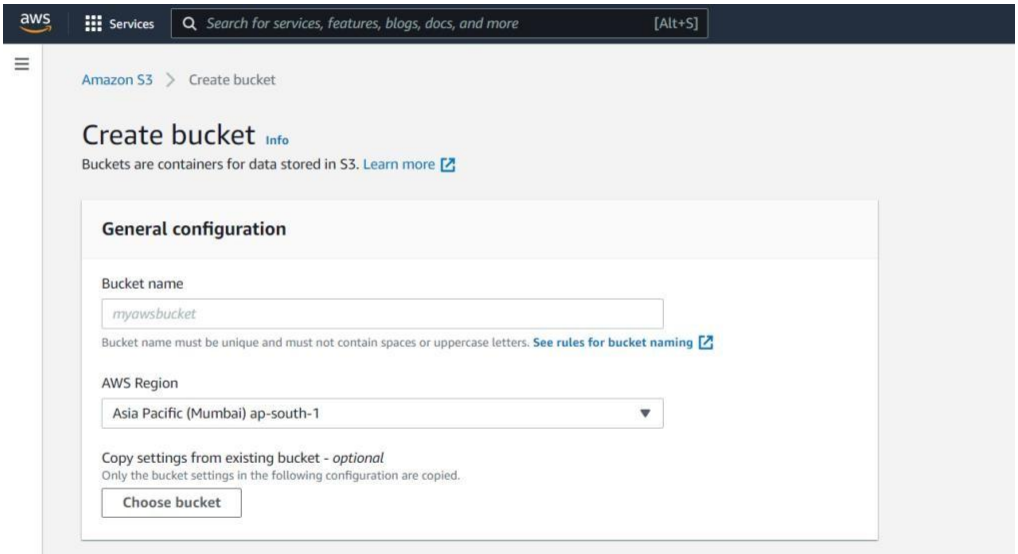
1. Login to AWS console and access S3 service



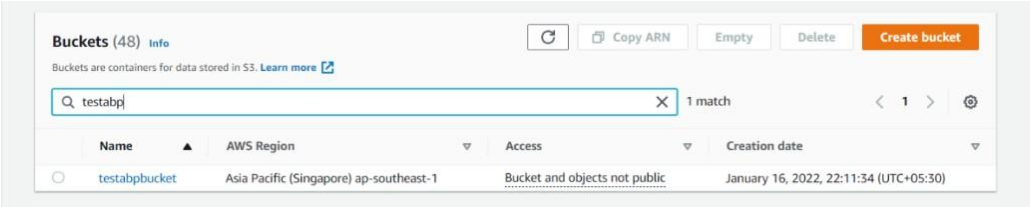
2. Create a bucket



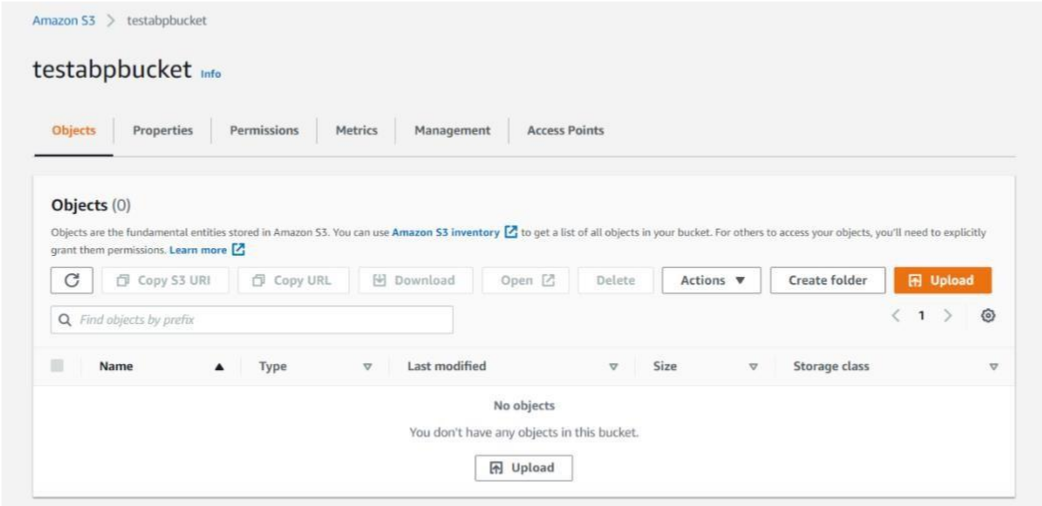
3. Add details of bucket (Note choose a unique name for your bucket)



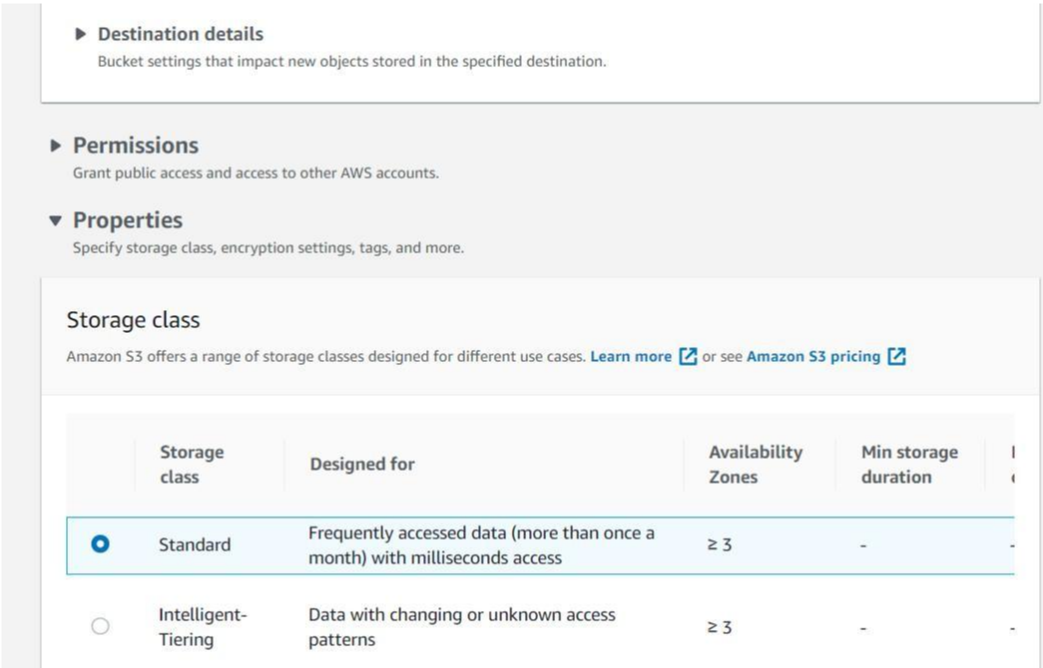
4. Search for bucket name



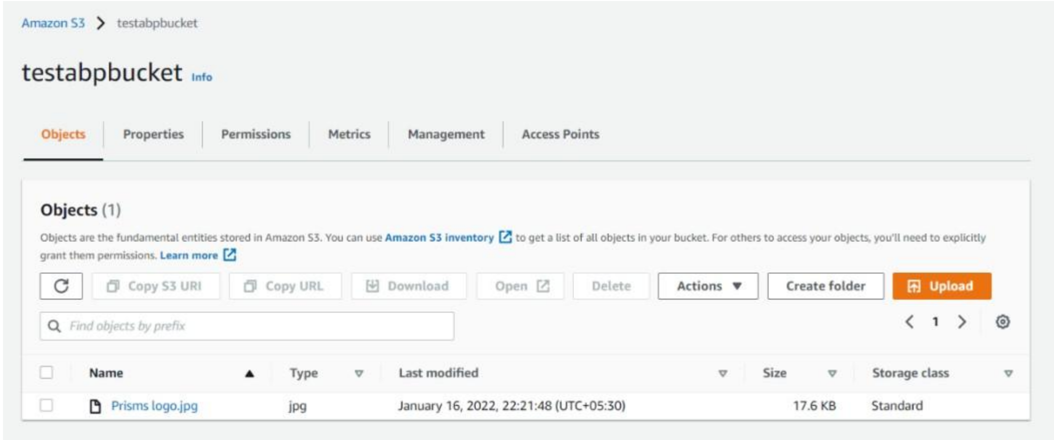
5. Click on bucket name for adding objects(items) inside buckets



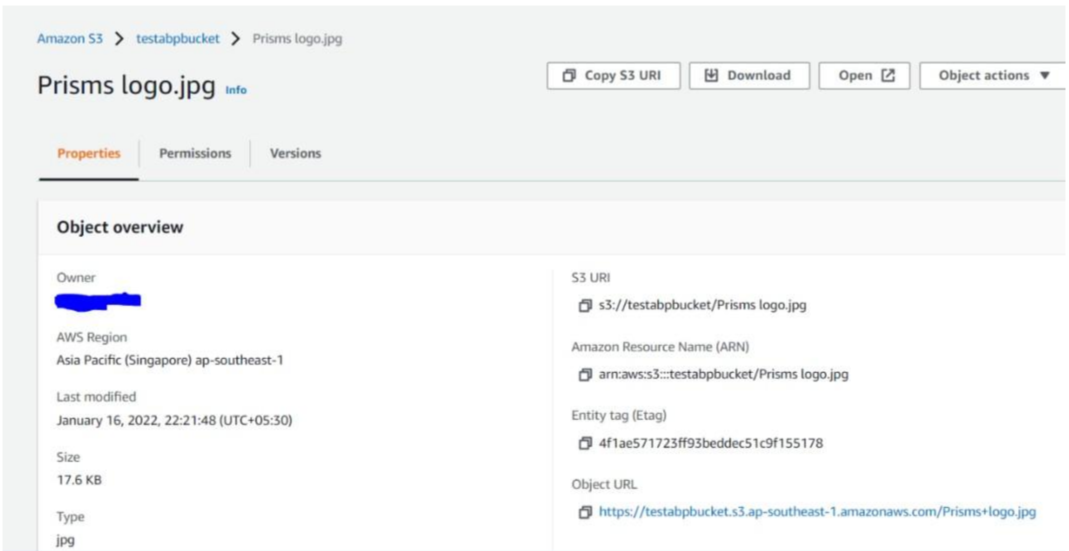
6. Select appropriate options while adding file. Same options can be selected later too.



7. Close window and go back to created bucket and open it. The object tab shows itemsin bucket



8. More details are shown by clicking on object name



9. Copy Object URL and paste in browser for accessing uploaded image
10. If error add necessary permissions to file.

Conclusion: S3 bucket is created successfully and objects are added to it. These objects can be accessed through Object URL. Also different types of S3 storage classes are observed.